



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VKG & BROTHERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

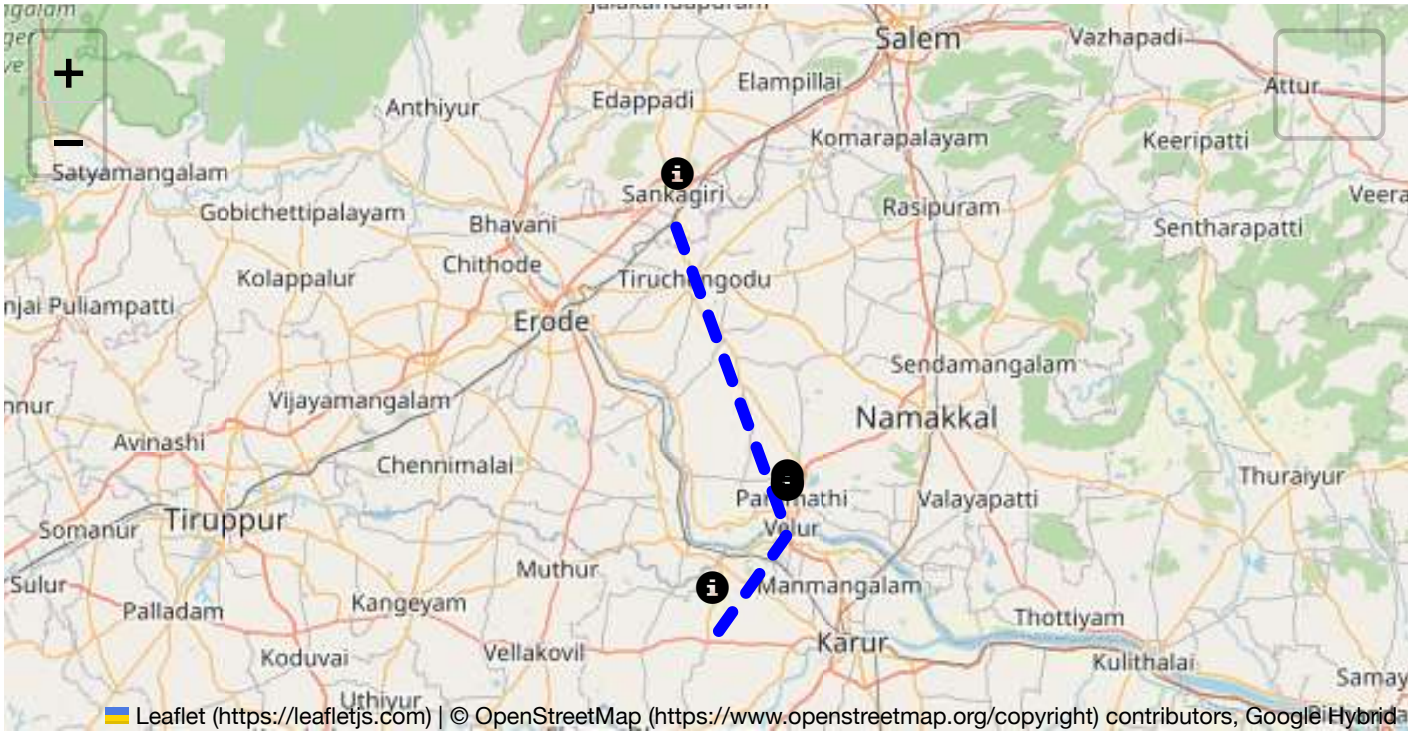
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

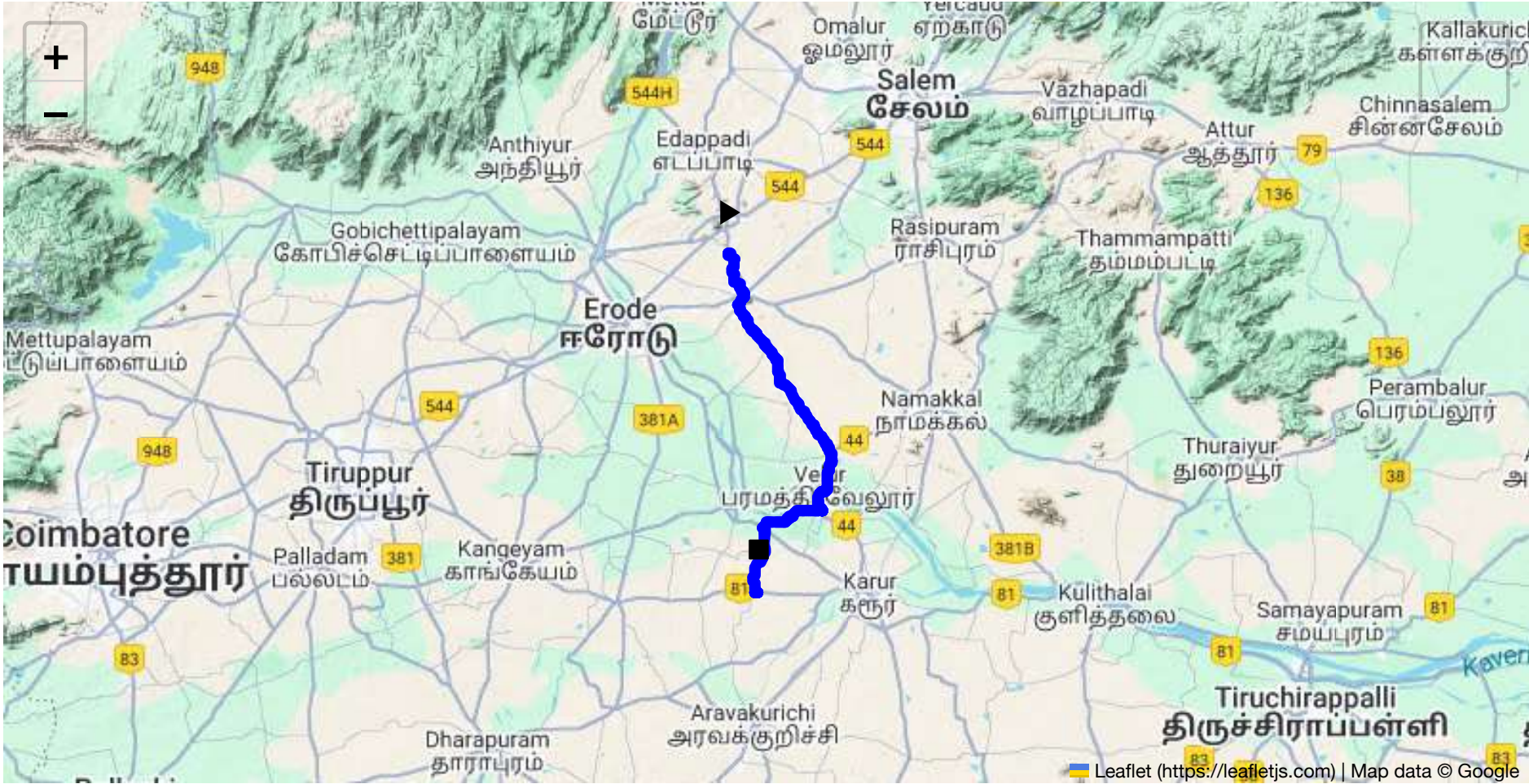
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 74.44 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours

Start: (11.4381, 77.8734)
End: (10.958783, 77.913729)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route begins in Sangagiri, Tamil Nadu, and follows a series of highways and bypasses to reach Aariyur Rd in K. Paramathi, Tamil Nadu. It passes through key waypoints including Nanjai Pugalur and Velayuthampalayam. The total distance is approximately 74.44 kilometers.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- Weather Conditions:** The region typically experiences a tropical climate with hot summers, heavy monsoon rains, and mild winters.
- Weather-Related Hazards:** Monsoon season (June to September) can cause heavy rain, leading to flooding and road damage. Fog can be an issue during winter mornings, reducing visibility.

3. Analysis of Traffic Patterns

- Peak Hours:** Traffic is generally heavier during morning (8 AM to 10 AM) and evening (5 PM to 8 PM) due to local commuting.
- Congestion-Prone Areas:** Notable congestion may occur at Salem - Madurai Rd near Nanjai Pugalur and Velayuthampalayam bypass, with additional congestion likely in towns and near marketplaces.

4. Assessment of Road Quality and Infrastructure

- Road Quality:** The major highways are generally in good condition but secondary roads may have potholes or uneven surfaces.
- Infrastructure:** Adequate signage is present, but street lighting is limited in rural segments, creating potential hazards at night.

5. Suggestions for Alternative Routes for Emergencies

- Use closer bypasses and service roads around urban centers to bypass major congestion points.
- For severe weather, consider redirecting through Salem - Namakkal - Karur as an alternative longer route.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Trucks must adhere to specific lane use rules and speed limits.
- Movement of hazardous materials is restricted during certain hours in densely populated areas.
- Ensure proper documentation and placarding of hazardous materials as per Tamil Nadu transport regulations.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- There have been occasional incidents involving overturned trucks, mainly attributed to high speeds or adverse weather conditions.
- Past hazardous material incidents often linked to inadequate sealing of containers and driver inattentiveness.

8. Environmental Considerations and Sensitive Areas

- The route passes near agricultural areas; care must be taken to avoid spills that could contaminate the soil.
- Velayuthampalayam area is near water bodies, necessitating careful handling of hazardous materials to avoid water source contamination.

9. Analysis of Communication Coverage

- **Coverage:** Generally reliable along the highways, but potential dead zones exist in rural patches, particularly around Velayuthampalayam bypass.
- Drivers should ensure mobile devices have sufficient battery life and emergency contact lists are ready offline.

10. Estimated Emergency Response Times

- In urban areas like Sangagiri and K. Paramathi: Approximately 30-45 minutes.
- Rural segments such as Velayuthampalayam: 45-60 minutes due to longer distances for emergency responders.

12. Overall Summary of Risk Assessment

- **Risks:** Major risks include weather-related disruptions, road quality issues on secondary roads, and congestion in market areas.
- **Mitigation:** Equip drivers with real-time weather and traffic data, ensure vehicles are in prime condition, and maintain communication equipment.

- **Preparedness:** Training on handling hazardous materials and communication protocols is vital, alongside understanding of emergency routes and procedures.

This overall analysis indicates that while the route is generally manageable for heavy vehicles carrying hazardous materials, attention must be focused on weather conditions, road quality, and timely response to emergency situations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	11.07495, 78.00165	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	High	11.37862, 77.89488	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16764, 78.01565	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14480, 78.02291	15 KM/Hr
11	Turn	Medium	11.09026, 78.00252	30 KM/Hr
12	Turn	Medium	11.08187, 78.00379	30 KM/Hr
13	Turn	High	11.07466, 78.00842	15 KM/Hr
14	Blind Spot	Blind Spot	11.07466, 78.00808	10 KM/Hr
15	Turn	High	11.08580, 78.00283	15 KM/Hr
16	Blind Spot	Blind Spot	11.08587, 78.00323	10 KM/Hr
17	Turn	Medium	11.08198, 78.00373	30 KM/Hr
18	Turn	High	11.07467, 78.00842	15 KM/Hr
19	Turn	High	11.07466, 78.00808	15 KM/Hr
20	Blind Spot	Blind Spot	11.08205, 78.00354	10 KM/Hr
21	Turn	High	11.08195, 78.00343	15 KM/Hr
22	Turn	High	11.08035, 78.00356	15 KM/Hr
23	Turn	High	11.07982, 78.00194	15 KM/Hr
24	Turn	Medium	11.07586, 77.97536	30 KM/Hr
25	Turn	Medium	11.06102, 77.93084	30 KM/Hr
26	Turn	Medium	11.06096, 77.93064	30 KM/Hr
27	Turn	Medium	11.05170, 77.92440	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
28	Turn	Medium	11.05118, 77.92452	30 KM/Hr
29	Turn	High	11.05061, 77.92520	15 KM/Hr
30	Turn	High	10.95905, 77.91206	15 KM/Hr
31	Turn	Medium	10.95889, 77.91232	30 KM/Hr
32	Turn	Medium	10.95988, 77.91591	30 KM/Hr
33	Blind Spot	Blind Spot	10.96011, 77.91611	10 KM/Hr
34	Turn	High	10.96001, 77.91614	15 KM/Hr
35	Turn	High	10.95907, 77.91387	15 KM/Hr
36	Turn	High	10.95883, 77.91393	15 KM/Hr
37	Turn	High	10.95877, 77.91380	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Velauthapalayam	11.0729521, 78.0023123	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.07495, 78.00165



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.09026, 78.00252



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.08187, 78.00379



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.07466, 78.00842



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.07466, 78.00808



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.08580, 78.00283



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.08198, 78.00373



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.07467, 78.00842



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.07466, 78.00808



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.08205, 78.00354



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.08195, 78.00343



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.08035, 78.00356



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.07982, 78.00194



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.07586, 77.97536



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.06102, 77.93084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.06096, 77.93064



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.05170, 77.92440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.05118, 77.92452



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.05061, 77.92520



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.95905, 77.91206



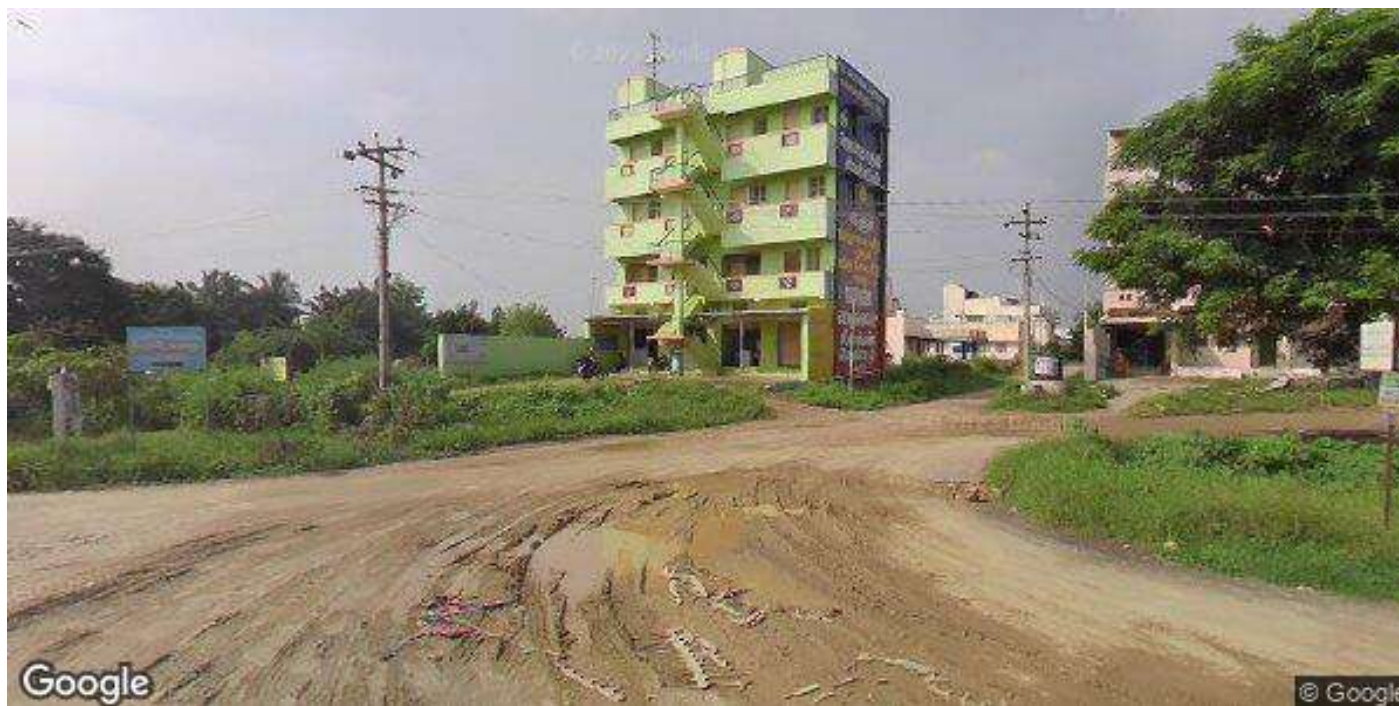
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.95889, 77.91232



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.95988, 77.91591



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.96011, 77.91611



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.96001, 77.91614



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.95907, 77.91387



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.95883, 77.91393



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.95877, 77.91380